LifeLine — Metrics and Tracking Plan v2 (FINAL)

LifeLine Logo

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Personal Safety & Check-In Group 39 · Phoenix Cohort · TS Academy PM Capstone Document version: 2.0 (FINAL — supersedes earlier metrics drafts) **Companion to:** *LifeLine PRD v3*, *LifeLine One-Pager v3* **Last updated:** 21 May 2026

1. Purpose

This plan defines what LifeLine measures, why, how, and what a production dashboard would look like. It satisfies the capstone deliverable “Metrics & tracking plan (North Star + 3 supporting + event tracking)” and goes further by specifying V2+ instrumentation when real backend telemetry becomes possible.

The MVP itself does not yet emit telemetry — it is a static client-side prototype. This plan documents (a) what events the MVP *would* emit if instrumented, (b) the full event taxonomy for V2 onwards, and (c) the dashboard a weekly product review would consult.

2. North Star metric

2.1 MVP North Star (current)

% of journeys ending in successful safe-arrival confirmation.

Formula: (count of Safe Arrival trips where 'I've arrived safely' was tapped before ETA expired) / (count of Safe Arrival trips started) × 100

Target: ≥ 90% at MVP review; ≥ 95% as the product matures.

Why this is the right North Star for MVP:

- Directly tied to LifeLine’s core promise — “safety in one tap”
- Observable from launch (a journey either ends with confirmation or it doesn’t)

- Captures both user behaviour and product reliability
- Demo-able at the capstone presentation
- Cannot be gamed by superficial engagement

2.2 Long-term North Star evolution (V2+)

> **Note for both judges and team:** Once LifeLine accumulates sustained usage data (≥ 3 months post-V2 launch), the North Star will evolve to:
>
> **Successful Safety Interactions Per Active User per week.**
>
> *Formula:* (safe-arrival confirmations + meaningful SOS activations + completed scheduled check-ins) / weekly active users

Why this evolution makes sense:

- Treats both flows (Safe Arrival and SOS) as legitimate sources of product value
- Captures engagement depth, not just completion of one flow
- Harder to game — high numbers require real, sustained use
- A better measure of the product fulfilling its mission across diverse user situations

Why this is not the MVP North Star:

- Requires longitudinal user data MVP does not produce
- Requires defining “meaningful SOS activation” (excluding accidental triggers) — which itself needs production data to calibrate
- Not directly demo-able on capstone day

2.3 Definition of success (compound threshold)

This is a compound *threshold*, not a continuously-tracked metric. It defines whether the MVP “passes” at production launch.

The MVP is considered successful at *production launch* (V2+) if **all** the following hold simultaneously:

- Safe-arrival check-in completion ≥ 90%
- Day-30 user retention ≥ 40%
- Alert dispatch latency < 5 seconds
- False-alert rate < 10%

3. Supporting metrics

The North Star answers “is the product working?” Supporting metrics answer “*why*?” Each explains a different mechanism behind the headline.

3.1 Activation rate

Definition: % of new users completing onboarding (OTP verified) AND adding ≥ 1 trusted contact, within 24 hours of first install. **Target:** 70% **Why it matters:** A user with zero contacts cannot use either core flow. Activation is the gate that turns a download into a usable account. **Diagnostic value if drops:** Onboarding friction, OTP failure, unclear value proposition.

3.2 Daily habit (frequency)

Definition: Safe Arrival sessions per active user per week. **Target:** ≥ 3 per active user per week **Why it matters:** Value compounds with use; the more journeys routed through Safe Arrival, the more the trusted-contact relationship becomes a real safety net. **Diagnostic value if drops:** Users forgetting the app; friction of starting a trip too high.

3.3 Emergency effectiveness

Definition: % of SOS activations where at least one trusted contact responds (via app or direct call) within 2 minutes. **Target:** 80% **Why it matters:** An SOS no one responds to is worse than no SOS — creates false sense of security. **Diagnostic value if drops:** Contacts not reachable, contacts not understanding what to do, notification reliability problems.

3.4 False-positive rate

Definition: % of SOS triggers cancelled within the 2:45 escalation window. **Target:** < 15% **Why it matters:** High false-positive suggests accidental triggers (UX issue) or user mistrust of cancel. **Diagnostic value if rises:** Pocket-bump activations, UX confusion about cancellation, test triggers miscounted.

4. Extended product metrics framework (V2+ tracking)

For when full instrumentation is available, the following framework guides ongoing measurement.

4.1 Acquisition

Metric	Definition	Target
App downloads	Total installs	5K+ MVP installs
Signup conversion rate	% of installs completing registration	> 60%
OTP verification success	% verifying OTP	> 95%
Cost per acquisition	Cost per active user	Optimise
Referral invite rate	% inviting trusted contacts	> 60%

4.2 Activation

Metric	Definition	Target
Trusted contact setup rate	% adding ≥ 1 contact	> 80%
First alert completion rate	% sending first alert/check-in	> 75%
Time to first safety action	Time before first action	< 10 min
Onboarding completion	Users finishing onboarding	> 60%
First live location share	Users sharing ETA/location	> 60%

4.3 Engagement

Metric	Definition	Target
DAU	Daily active users	Growth trend
WAU	Weekly active users	60–70% engagement
DAU/WAU ratio	Stickiness	> 40%
Avg check-ins per user	Weekly check-ins	Increasing trend
Session frequency	Avg opens per week	> 3

4.4 Retention

Metric	Definition	Target
Day 1 retention	Return after first day	> 50%
Day 7 retention	Return after 7 days	> 35%
Day 30 retention	Return after 30 days	> 40% (MVP goal)
Churn rate	Users becoming inactive	Minimise
Re-engagement rate	Returning after inactivity	Improve over time

4.5 Safety & reliability (most critical)

Metric	Definition	Target
Alert delivery time	Trigger → recipient delivery	< 5s
Notification success rate	Push notifications delivered	≥ 99.5%
Escalation success rate	Alerts acknowledged during escalation	> 90%
False alert rate	Accidental activations	< 10%
Avg acknowledgment time	Contact response speed	< 2 min
SMS fallback success	Fallback deliveries	> 95%
GPS accuracy rate	Accurate location delivery	> 95%

4.6 Technical performance

Metric	Definition	Target
App launch time	Cold start	< 3s
API response time	Backend latency	< 500ms
Crash-free sessions	App stability	> 99%
Battery consumption	Battery impact during tracking	Low
GPS refresh accuracy	Location update consistency	5–10s
Backend uptime	Service availability	99.9%

4.7 Privacy & trust

Metric	Why it matters
Permission acceptance rate	Trust in location/privacy prompts
Privacy settings usage	User awareness/control
Account deletion rate	Potential dissatisfaction/trust issues
Consent revocation rate	Privacy concerns

Metric	Why it matters
Security incident count	Must remain near zero

5. Event tracking plan

The event taxonomy below is what LifeLine emits from V2 onwards. Naming convention: flow_action_object — lowercase, underscored.

5.1 Authentication events

Event name	Trigger	Key properties
signup_started	User opens signup	device_type, network_type
signup_completed	Registration successful	total_time_seconds
otp_sent	OTP dispatched	phone_country_code
otp_entered	User submits OTP	attempt_number, correct
otp_verified	Correct OTP entered	attempts_taken
otp_max_attempts_reached	User exhausts 3 attempts	
login_success	User logs in	
login_failed	Failed login attempt	failure_reason

5.2 Safe Arrival events

Event name	Trigger	Key properties
safe_arrival_trip_started	User taps “Share with contacts”	trip_id, destination, eta_minutes, contacts_count, location_permission_status
safe_arrival_location_granted	User grants location permission	trip_id
safe_arrival_location_denied	User denies location	trip_id, manual_address_provided
safe_arrival_manual_address_entered	User types address after denying	trip_id
safe_arrival_tracking_started	Live tracking screen loads	trip_id, gps_accuracy_metres
safe_arrival_confirmed	User taps “I’ve arrived safely”	trip_id, time_to_confirm_seconds
safe_arrival_overdue	ETA countdown reaches 0:00 without confirmation	trip_id, overdue_at_timestamp
safe_arrival_contacts_auto_notified	Contacts auto-	trip_id,

Event name	Trigger	Key properties
	notified due to overdue	contacts_notified
safe_arrival_post_overdue_confirmed	User confirms safe after overdue	trip_id, minutes_post_overdue
safe_arrival_trip_cancelled	User cancels trip manually	trip_id, minutes_into_trip

5.3 SOS events

Event name	Trigger	Key properties
sos_hold_started	User starts holding SOS	start_timestamp
sos_hold_aborted	User releases before 3 seconds	hold_duration_ms
sos_activated	3-second hold completes	sos_id, gps_lat, gps_lng, contacts_notified
sos_cancelled	User taps "I'm safe — cancel"	sos_id, seconds_remaining, was_test
sos_escalated	2:45 timer expires without cancel	sos_id
sos_contact_acknowledged	Contact opens alert	sos_id, contact_id, response_time_seconds
sos_contact_responded	Contact takes action	sos_id, contact_id, action_type
sos_resolved	SOS marked resolved	sos_id, resolution_path, total_duration_seconds
shake_detected	Device motion exceeds threshold	toast_shown

5.4 Location & tracking events

Event name	Trigger	Key properties
location_shared	Live location started	session_id
eta_shared	ETA sent	session_id, eta_minutes
tracking_session_started	Tracking begins	session_id
tracking_session_ended	Tracking ends	session_id, ended_reason
gps_permission_denied	GPS access denied	

5.5 Check-in events (V3+)

Event name	Trigger	Key properties
checkin_created	Scheduled check-in set	checkin_id, scheduled_at
checkin_completed	User confirms safe	checkin_id, minutes_early_or_late
checkin_missed	User misses check-in	checkin_id
reminder_sent	Reminder notification triggered	checkin_id

5.6 Notification events

Event name	Trigger	Key properties
push_sent	Notification dispatched	notification_id, type
push_delivered	Push received	notification_id, latency_ms
push_opened	User taps notification	notification_id
sms_fallback_triggered	SMS backup activated	reason

5.7 Contact management events

Event name	Trigger	Key properties
contact_added	User adds a trusted contact	contact_count_after, consent_status
contact_removed	User removes a contact	contact_count_after
contact_invitation_sent	Invitation SMS sent to contact (V2)	contact_id
contact_invitation_accepted	Contact accepts invitation (V2)	contact_id, time_to_accept

5.8 Failure path events

Event name	Trigger	Key properties
network_unavailable_during_action	Network drop during action	action_name, queued_for_retry
map_tile_load_failed	OSM tile load error	tile_count_failed
app_crashed	Unhandled exception (V2 integrates Sentry)	error_message, screen_name

6. Dashboard specification (mocked)

A weekly product review would consult a dashboard with the following structure. For the capstone, this is documented as a wireframe. V2 will build it.

6.1 Dashboard layout — text wireframe

LIFELINE — PRODUCT DASHBOARD Week 23 · May 2026

NORTH STAR	ACTIVATION RATE	SAFE ARRIVAL / WAU
% Safe Arrival Confirmed	72% ↑	3.4 / wk ↑
92.4% ↑	+2pp vs LW	↑ 0.2 vs LW
Target: ≥ 90%	Target: 70%	Target: ≥ 3

EMERGENCY EFFECTIVENESS	FALSE POSITIVE RATE	NOTIFICATION DELIVERY
84% ↑	12% ↓	99.7% →
Target: 80%	Target: <15%	Target: ≥ 99.5%

FUNNEL VIEW (THIS WEEK)

App opens	100%	1,243
OTP verified	85%	1,057
First contact	72%	895
First trip	55%	684
First confirmation	50%	621
Returning week 2	35%	435

COHORT RETENTION

Day 1: 85%
Day 7: 42%
Day 14: 38%
Day 30: 41%

RECENT SOS EVENT LOG

#SOS-2389	20 May 21:14	Resolved via contact	2m 14s
#SOS-2388	20 May 19:02	Cancelled by user	0m 18s
#SOS-2387	19 May 22:47	Resolved via contact	1m 56s
#SOS-2386	19 May 18:33	Cancelled by user	0m 09s (test)

🚨 ALERTS

✓ All metrics within thresholds. No alerts active.

6.2 Dashboard sections explained

1. **Top row (4 tiles):** North Star + 3 supporting metrics with week-over-week sparkline arrows
2. **Second row (3 tiles):** Diagnostic numbers — emergency effectiveness, false-positive rate, notification delivery
3. **Funnel view:** Weekly user progression from open → retain
4. **Cohort retention:** Curve showing Day 1, 7, 14, 30 retention per cohort
5. **Recent SOS event log:** Privacy-respecting (hashed IDs) log for the safety team to review patterns
6. **Alerts block:** Red-status panel that appears only on threshold breaches:
   - False-positive rate > 20% in past 7 days
   - Emergency effectiveness < 60% in past 7 days
   - Notification delivery success < 95% in past 24 hours
   - Any SOS unresolved for > 24 hours

7. Instrumentation approach

7.1 MVP (current)

The MVP is a single-file static HTML app with no backend. It does not currently emit any events. This is acknowledged as a capstone gap and is mitigated by:

- Manual test results documenting 30/30 passing test cases
- Build evidence describing what events the app *would* emit
- This Tracking Plan defining V2 instrumentation in full

7.2 V2 (planned)

When LifeLine moves to Supabase in V2, instrumentation is added as follows:

1. **Event capture:** Lightweight in-app emitter batches events; posts to a Supabase Edge Function endpoint every 30 seconds or on app background
2. **Event storage:** Events land in lifeline_events Postgres table with schema (event_id uuid, user_id uuid, event_name text, properties jsonb, occurred_at timestamptz, server_received_at timestamptz)
3. **Privacy by design:** No raw GPS in events; only derived properties like gps_accuracy_metres, distance_from_home_metres_bucketed
4. **PII handling:** No phone numbers or addresses in events; only opaque IDs
5. **Retention:** Event data retained 90 days; aggregated metrics retained indefinitely

7.3 Tooling stack

Need	Tool
Event capture and storage	Supabase (matches the rest of the stack)

Need	Tool
Product analytics	Firebase Analytics or Mixpanel
Crash monitoring	Firebase Crashlytics or Sentry
Performance monitoring	Firebase Performance
Notifications	Firebase Cloud Messaging
Session replay	PostHog or LogRocket
Infrastructure monitoring	AWS CloudWatch
Database monitoring	Supabase console / PostgreSQL monitoring
Dashboard visualisation	Metabase or Grafana on top of Supabase Postgres
Alerting	Supabase Edge Functions emitting to email/Slack on threshold breach

8. Privacy considerations

A safety app collects sensitive data. The tracking plan is designed with these constraints:

- **No raw GPS in event logs.** Location is approximated to neighbourhood-level buckets for analytics.
- **No phone numbers or names in event properties.** Only opaque user_ids and contact_ids.
- **No event tracking can be tied back to a real-life identity** without account-level access (which is restricted).
- **Users can delete their account and all associated event history** via in-app deletion (V2 feature).
- **Aggregate metrics** are computed and stored separately from individual events.

9. How metrics drive decisions

If this metric moves...	The team should...
Activation rate drops below 60%	Audit onboarding for friction; A/B test a shorter flow
Daily habit drops below 2 sessions/week	Add gentle reminder nudges; investigate forgetting
Emergency effectiveness drops below 70%	Audit contact onboarding (reachable channels?); mandate test alert at onboarding
False-positive rate rises above 20%	Review SOS UX; consider extending cancel window
Safe-arrival confirmation rate drops below 85%	Audit "I've arrived safely" button placement; consider auto-confirm -on-geofence
Notification delivery success	Audit SMS gateway health; trigger

If this metric moves…	The team should…
drops	redundancy

10. Quarterly metric review cadence

- **Weekly:** PM lead reviews dashboard top row; flags alerts to team
- **Monthly:** Full team reviews funnel and cohort retention; agrees one experiment
- **Quarterly:** Strategy review of North Star — consider whether to evolve from MVP NSM to long-term NSM
- **Annually:** Reset targets based on prior year baseline

11. Known gaps and next steps

1. **No live telemetry in MVP** — addressed in V2 instrumentation plan (Section 7.2)
2. **"Meaningful SOS activation" not yet operationalised** — will be calibrated with first 3 months of V2 production data
3. **No A/B testing infrastructure in V2 plan** — recommended add for V3
4. **Privacy review of event taxonomy** — needs sign-off from a privacy reviewer (in real production) before V2 launch

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